University of Bolton

MODULE DATABASE

2025-26 > Onwards

COM7401 (Validated in 2025-26)

Research Methods

Credits	20
Level	HE7
Type	Standard
Trimester 3?	Yes
ECTS	10
Marking Scheme	Numeric mark scheme (i.e. mark of 0-100)
Pass Mark	50%
Delivery Type	This Module requires you to attend particular classes or events at particular times and in particular locations.
Module Outline	This module develops advanced research knowledge and skills essential for postgraduate study in computing and related disciplines. It enables students to design, conduct, and evaluate ethical and methodologically sound research while developing confidence as independent investigators. The module draws on contemporary research practices and debates to ensure relevance to today's technological and professional contexts. Flipped learning is used to augment engagement, allowing students to explore preparatory materials before in-class workshops focused on application and discussion. The module complements and augments all others in the programme by strengthening students' ability to apply critical thinking, research design, and evidence-based analysis across diverse contexts. Through an applied portfolio, students integrate theoretical understanding with practical research design, consolidating learning through reflection, ethics, and professional communication.

Indicative Content

1 Foundations of research philosophy and design

2	Quantitative, qualitative, and mixed-methods approaches
3	Sampling and data collection strategies
4	Data analysis and interpretation techniques
5	Ethics, governance, and professional conduct in research
6	Academic writing and dissemination of findings
7	Use of technology in research (e.g., data analysis tools, referencing software, AI-assisted methods)
8	Reflection, evaluation, and the assessment of research quality

Learning Outcomes

On successful completion of this Module you will be expected to be able to:

1	Critically evaluate a range of research methodologies and philosophical approaches relevant to computing and data-oriented disciplines.
2	Design and justify an ethical and feasible research proposal informed by critical engagement with relevant literature.
3	Synthesise research findings through appropriate analysis, interpretation, and academic communication.

Learning And Teaching Strategy

Learning and teaching combine lectures, seminars, and workshops designed to develop advanced understanding of research principles, methodologies, and academic practice. Students will engage critically with both academic and industry sources to explore how research informs and advances computing disciplines. Flipped learning is used to support engagement with preparatory materials, enabling class time to focus on analysis, discussion, and the application of methods to practical and ethical challenges. Workshops provide opportunities to design and evaluate research proposals, while peer review and group discussion encourage reflection and critical thinking. Independent study allows students to deepen their understanding through literature exploration, data handling, and the development of methodological rigour.

Learning & Teaching Methods

Method	KIS	Hours
Scheduled	Scheduled	45

Independent	Independent	155
Total		200

Continuous Assessment Strategy

Continuous assessment underpins learning throughout the module, ensuring sustained engagement with research design and development. Formative activities, such as proposal outlines, literature critiques, and feedback workshops, enable students to refine their research ideas and methods in response to tutor and peer feedback. These formative tasks contribute directly to the development of the final portfolio. The summative assessment consists of a portfolio of coursework, demonstrating the ability to design, justify, and communicate a research proposal supported by ethical awareness and critical reflection. This structure allows students to evidence academic integrity, professional communication, and readiness for the subsequent project module.

Summative Assessments

	Assessment	KIS	Description		Learning Outcomes	Marking Scheme	Passmark	KIS Weighting
001	Portfolio	Coursework	A portfolio culminating in a report detailing the research background, critical review of relevant literature, and proposed methodology, supported by an oral presentation that discusses and defends the proposed investigation within a computing context.	FINAL	1 2 3	Standard University Regulations	-	100%

Learning Resources

Essential Reading

Cohen, L., Manion, L. and Morrison, K. (2018) Research Methods in Education (8th ed.). Milton Park: Routledge, Taylor & Francis Group.

Creswell, J. W. and Creswell, J. D. (2022) Research Design: Qualitative, Quantitative, and Mixed Methods Approaches (6th ed.). Thousand Oaks: SAGE Publications.

Gray, D. E. (2025) Doing Research in the Real World (6th ed.). London: SAGE Publications.

Hammond, M. and Wellington, J. (2020) Research Methods: The Key Concepts (2nd ed.). Milton Park: Routledge, Taylor & Francis Group.

Recommended Reading

Field, A. P. (2022) An Adventure in Statistics: The Reality Enigma (2nd ed.). Thousand Oaks: SAGE Publications.

Hammond, M. and Wellington, J. (2020) Research Methods: The Key Concepts (2nd ed.). Milton Park: Routledge, Taylor & Francis Group.

Mbanaso, U. M., Abrahams, L. and Okafor, K. C. (2023) Research Techniques for Computer Science, Information Systems and Cybersecurity. Berlin: Springer Nature.

Talis Aspire Reading lists

Additional Teaching, Learning and Assessment arrangements for asynchronous delivery of this module only

Feedback to Students	Formative and summative assessment feedback will be made available verbally and/or in written form. Feedback will be provided within the terms of the University's guidelines which are provided in the Module Guides.

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